DiagnoTrace: A Holistic, Explainable Patient View

One longitudinal view of each patient, across doctors and time, where the AI never has the last word.

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Problem

Physicians assess a patient mainly from the examination during the visit, what the patient reports, and their own practice's past records. Live data between visits (e.g. from wearables) is missing, precise findings from other doctors rarely arrive, and Germany's electronic patient record (ePA) mostly holds documents. Many decision-support tools are rule sets or black boxes that rarely explain warnings, ignore free text and need core changes per source. Slow trends can be missed, delaying diagnosis.

Idea

DiagnoTrace merges everything known about a patient into one timeline and checks it continuously. An event-driven plug-in design lets each new fact trigger independent checks, and new sources or checks need no core change (**extensibility**). All data is stored in fixed types (e.g. lab value, medication, hospital stay) that record its origin, so every output traces back to its source (**explainability**).

- **Data intake** reads mock ePA documents, a wearable feed and uploaded blood reports. An LLM turns free text into the same fixed types.
- **Checks** are plug-ins: simple rules and basic ML risk models.
- **AI chat** summarises the patient, suggests diagnoses and answers the doctor's questions, citing only stored facts and open-licensed WHO guidelines. It cannot raise or suppress alerts.
- **Web app** shows the timeline, alerts and chat; the doctor decides.

Novel: live data and free-text records from all doctors in one view the doctor can question, with rules and models raising alerts and the AI explaining them. **Trade-offs:** fixed types vs. unforeseen data; LLM extraction vs. fixed parsers; event bus vs. direct calls. **Out of scope:** real patient data (special-category data under GDPR), real ePA integration, clinical validity.

Benefits

- **Holistic view** – addresses single-visit snapshots; shown by one timeline merging three sources.
- **Traceable answers** – addresses black-box advice; target $\geq$95% of AI statements correctly cited.
- **Reliable extraction** – addresses unused free text; target $\geq$90% precision and recall.
- **Safe AI use** – addresses wrong AI output; target 0 alerts raised or suppressed by the chat.
- **Easy extension** – addresses costly integration; target 0 core files changed per plug-in.

Action

1. Specify requirements, quality scenarios and fact types; document the architecture.
2. Build the core: event bus, fact store with audit log, plug-in registry.
3. **Minimum deliverable:** ePA extraction, wearables, rule checks, alerts, timeline.
4. Extend: blood-report upload, ML models, AI chat about the patient.
5. Evaluate the quality scenarios, prepare the presentation; two weeks buffer.

Python, Flask, Pydantic, PostgreSQL, OpenRouter, Docker Compose; synthetic data only. **Team** (one lead each, set at kickoff): core; data intake; extraction; checks; AI chat; web app.

AI usage statement. Claude Sonnet 5.5 (claude.ai): brainstorming, review. Claude Opus 5.5 (Claude Code): drafting, fact-checking, typesetting. The team reviewed and adapted all content.